

- 3 The tuatara, *Sphenodon punctatus*, is a reptile that is native to New Zealand and is found in the wild only in that country. Fossil evidence shows that there has been little morphological change in the tuatara over the last 200 million years.

Figure 3.1 shows a tuatara.



Figure 3.1

- (a) The tuatara inhabited the main islands and the smaller islands of New Zealand for most of the last 200 million years. Around 800 years ago, humans arrived in New Zealand. In the last hundred years, a rapid decrease in the number of tuatara has occurred and now they are found in the wild on only 32 small islands.

Suggest why the number of tuatara has decreased in most parts of New Zealand.

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(b) Moutoki Island is one of the small islands where there is a wild population of tuatara. The size of the population of tuatara was estimated using the mark–release–recapture method:

- A sample of 23 tuatara was captured using a humane (harmless) trap.
- The tuatara were marked in a way that did not harm them.
- The marked tuatara were released into the area where they were captured.
- A period of time was allowed for the marked tuatara to mix into the population.
- A second sample of 38 tuatara was captured, in which 6 individuals were marked.

Use the Lincoln Index to calculate the size of the tuatara population on Moutoki Island.

Give your answer to the nearest whole number.

Lincoln Index

$$N = \frac{n_1 \times n_2}{m_2}$$

Key to symbols:

N = estimate of population size

n_1 = number of individuals captured in first sample

n_2 = number of individuals (both marked and unmarked) captured in second sample

m_2 = number of marked individuals recaptured in second sample

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(c) The optimum body temperature for the tuatara is 16–21 °C, the lowest of any reptile. The low body temperature results in a slow metabolism and therefore slow rate of respiration.

Suggest **and** explain why a low body temperature would result in a slow rate of respiration.

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(d) The tuatara belongs to the kingdom Animalia.

Outline the main characteristics of the kingdom Animalia.

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